Multi-Mechanism Pre-training Enables Robust Physics-Informed Fouling Prediction under Sparse Industrial Data

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Abstract

Heat exchanger fouling is a major source of energy loss in the process industry, yet accurate prediction of fouling resistance evolution $R_f(t)$ is hindered by severe data scarcity in industrial settings. This study proposes a multi-mechanism pre-training framework that leverages five distinct fouling ODE models to pre-train a Physics-Informed Neural Network (PINN) on diverse synthetic data, followed by parameter-efficient fine-tuning on extremely sparse real measurements ($N = 5$–30 points). We validate the framework on 18 real fouling curves digitized from three peer-reviewed studies, spanning crystallization fouling, crude oil fouling, and $CaCO_3$ precipitation with induction periods. A critical baseline reveals that direct ODE curve fitting fails on 15 of 18 curves with sparse data ($R^2 \approx 0$ or negative), while PINN pre-training rescues 12 of these failures. Key findings include: (1) multi-mechanism pre-training is essential not for coverage but for *diversity*—single-mechanism (Kern-Seaton only) pre-training catastrophically overfits, achieving $R^2 = -1329$ on its own target curve type, while multi-mechanism pre-training reaches $R^2 = 0.94$ on the same curve; (2) the framework achieves mean $R^2 > 0.90$ on 12 of 18 curves with only 15 training points in 5-seed multi-seed validation; (3) the pre-trained PINN consistently outperforms Random Forest in extrapolation ($\Delta R^2 = +1.3$ to $+18.7$ at $N = 5$); and (4) PINN-driven cleaning strategies reduce dimensionless cost ratios by 20–97% versus fixed-interval policies. The approach requires only publicly available physics models for pre-training and approximately 3 seconds of fine-tuning per heat exchanger on consumer GPU hardware.

Keywords: Physics-Informed Neural Networks, heat exchanger fouling, sparse data, transfer learning, cleaning optimization, multi-mechanism modeling

1. Introduction

1.1. Problem Background

Heat exchangers are ubiquitous in the chemical, petroleum, power generation, and food processing industries. During operation, dissolved salts ($CaCO_3$, $CaSO_4$), particulates, and organic compounds deposit on heat transfer surfaces, forming a fouling layer whose thermal conductivity (0.5–2 W/m·K) is one to two orders of magnitude lower than that of the metal tube wall (50–400 W/m·K) [1]. The resulting increase

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in overall thermal resistance forces plants to either increase driving temperature differences (and thus fuel consumption) or accept reduced throughput.

The fouling thermal resistance $R_f(t)$ evolves according to the net effect of deposition and removal processes:

$$\frac{dR_f}{dt} = \phi_d - \phi_r \tag{1}$$

where ϕ_d is the deposition rate and ϕ_r is the removal rate—both functions of local temperature, velocity, concentration, and surface conditions.

1.2. The Cleaning Decision Problem

Since fouling is unavoidable, industrial practice relies on periodic cleaning to restore heat transfer efficiency. The cleaning scheduling problem is an optimization: clean too frequently and incur excessive downtime costs; clean too infrequently and suffer escalating energy penalties. The optimal cleaning interval minimizes the sum:

$$\min_{t_{\text{clean}}} [\text{Energy Penalty}(t_{\text{clean}}) + \text{Cleaning Cost} + \text{Downtime Cost}] \tag{2}$$

This optimization requires accurate prediction of $R_f(t)$, specifically the time at which the fouling resistance will exceed a critical threshold $R_{f,\text{crit}}$. Without a reliable predictive model, cleaning decisions are made by heuristics—typically fixed calendar intervals—that are almost certainly suboptimal.

1.3. The Data Scarcity Problem

Accurate $R_f(t)$ prediction faces a fundamental obstacle: industrial fouling data is extremely scarce. Plant operators are reluctant to share operational data due to competitive sensitivity, and installing dedicated fouling monitoring instrumentation is expensive. The typical industrial scenario involves a handful of historical cleaning records and occasional temperature/pressure measurements convertible to R_f estimates.

Standard machine learning approaches (MLPs, random forests, gradient boosting) require hundreds to thousands of labeled examples to achieve acceptable accuracy. With only 5–30 data points—the realistic industrial scenario—purely data-driven methods either fail completely or produce physically implausible predictions.

1.4. PINNs as a Solution and Their Limitations

Physics-Informed Neural Networks (PINNs) [2] address data scarcity by augmenting the standard data loss with a physics-based regularization term:

$$\mathcal{L} = \underbrace{\text{MSE}(R_f^{\text{pred}} - R_f^{\text{obs}})}_{\text{Data Loss}} + \lambda \cdot \underbrace{\text{MSE}\left(\frac{dR_f}{dt} - (\phi_d - \phi_r), 0\right)}_{\text{Physics Loss}} \tag{3}$$

The physics loss enforces the governing ODE at collocation points where no observational data is available, effectively providing an infinite source of "free" training signal [3].

However, existing PINN approaches for fouling suffer from two critical limitations: (1) they use only the Kern-Seaton asymptotic model as the physics constraint, while real fouling exhibits diverse mechanisms—induction periods, threshold behavior, cleaning-induced discontinuities; and (2) all experiments to date have been conducted on synthetic data generated by the same model used in the physics loss, creating a self-consistency loop where the PINN is merely learning to invert a known equation rather than discovering genuine physical structure.

1.5. Contributions of This Work

This study addresses both limitations through a multi-mechanism pre-training framework. Our specific contributions are:

1. **Multi-mechanism ODE registry**: We implement five distinct fouling ODE models with a unified interface, enabling PINN pre-training on diverse synthetic fouling physics.
2. **Pre-training + fine-tuning pipeline**: We demonstrate that pre-training a PINN on multi-mechanism synthetic data creates a physics-informed prior that transfers to real fouling curves via lightweight fine-tuning on as few as 5–30 sparse data points.
3. **Comprehensive real-data validation**: We digitize and validate on 18 real fouling curves from three independent experimental studies, covering three fouling types and multiple surface conditions.
4. **Single vs. multi-mechanism ablation**: We establish that multi-mechanism diversity prevents catastrophic over-specialization—single-mechanism pre-training achieves $R^2 = -1329$ on its own target curve type, while multi-mechanism pre-training reaches $R^2 = 0.94$ on the same curve.
5. **ODE-fit baseline and honest assessment**: By comparing against direct ODE curve fitting, we show that pure physics models fail on 15 of 18 curves with sparse data, precisely identifying the regime where PINN pre-training provides value—and honestly acknowledging where simpler methods suffice.

2. Methodology

2.1. Multi-Mechanism Fouling ODE Registry

We implement five fouling kinetics models with a unified Python interface. Each model provides an analytical solution $R_f(t)$ (if available) for synthetic data generation and an ODE residual function $\mathcal{R}(dR_f/dt, R_f, \mathbf{p})$ for PINN physics loss computation.

Model 1—Kern-Seaton (KS): The classical asymptotic fouling model [4]:

$$\frac{dR_f}{dt} = A \exp\left(-\frac{E_a}{RT_w}\right) - B \cdot \tau_s \cdot R_f \tag{4}$$

$$R_f(t) = R_f^* \left(1 - \exp(-t/\tau)\right), \quad R_f^* = \frac{\phi_d}{B\tau_s}, \quad \tau = \frac{1}{B\tau_s}$$

Applicable to: crystallization fouling, particulate fouling with asymptotic behavior.

Model 2—KS with Initial Residual: Extends Model 1 with $R_f(0) = R_{f0} > 0$, modeling incomplete cleaning. Applicable to heat exchangers with residual fouling after chemical cleaning.

Model 3—Simplified KS (Phenomenological): A 3-parameter form for direct curve fitting:

$$\frac{dR_f}{dt} = \frac{R_f^* - R_f}{\tau}, \quad R_f(t) = R_{f0} + R_f^*(1 - e^{-t/\tau}) \tag{5}$$

Applicable to all asymptotic curves with unknown physical parameters.

Model 4—Induction + Linear Growth: Models the characteristic delay before rapid fouling onset:

$$R_f(t) = \begin{cases} 0 & t < t_{\text{ind}} \\ k_{\text{growth}} \cdot (t - t_{\text{ind}}) & t \geq t_{\text{ind}} \end{cases} \tag{6}$$

The analytical form is piecewise-defined and non-differentiable at $t = t_{\text{ind}}$. For PINN physics loss computation, we use a smoothed approximation with a sigmoid function (slope $\alpha = 10$) providing a continuous transition. The analytical solution used for data generation and curve fitting uses the exact piecewise form. Applicable to $CaCO_3$ crystallization on modified surfaces.

Model 5—Ebert-Panchal Threshold: Incorporates a critical shear stress below which fouling does not occur [5]:

$$\frac{dR_f}{dt} = \alpha \cdot \text{Re}^{\beta} \cdot \text{Pr}^{-0.66} \cdot \exp\left(-\frac{E_a}{RT_{\text{film}}}\right) - \gamma \cdot \tau_w \tag{7}$$

Applicable to crude oil fouling with threshold velocity/temperature conditions.

2.2. PINN Architecture

The PINN takes a unified 10-dimensional input vector:

$$\mathbf{x} = [t, T_w, v, p_1, p_2, p_3, p_4, p_5, p_6, \text{ode\_type\_id}]^T \tag{8}$$

where p_1–p_6 encode ODE-specific physical parameters and unused slots are zero-filled. The network has 4 hidden layers of 64 neurons each with tanh activation (13,249 parameters). The physics loss is computed per-trajectory using the appropriate ODE residual function from the registry.

2.3. Pre-training on Multi-Mechanism Synthetic Data

Pre-training data is generated from all five ODE models with stratified parameter sampling: In-domain (ID) parameters near typical phosphoric acid concentration conditions ($T_w = 340$–380 K, $v = 0.8$–2.5 m/s), Out-of-domain 1 (OOD-1) with shifted parameter ranges, and Out-of-domain 2 (OOD-2) with completely different fouling mechanisms. Each ODE model generates 200 ID + 100 OOD-1 + 100 OOD-2 trajectories (60 time points each, $\sigma = 2 \times 10^{-6}$ $m^2{\cdot}K/W$ Gaussian noise), yielding 2,000 total trajectories. Pre-training runs for 3,000 epochs with mini-batch sampling (8 trajectories per ODE type per epoch), physics loss weight $\lambda = 0.5$, and Adam optimizer with ReduceLROnPlateau scheduling.

2.4. Sparse Fine-tuning on Real Data

For each real fouling curve, the fine-tuning pipeline proceeds as:

1. **ODE model matching**: Select the appropriate ODE model based on the curve's visual characteristics and the fouling type reported in the source publication. Asymptotic curves (Source 001, Source 002) use Model 3; curves with visible induction periods (Source 003) use Model 4. In the current implementation, this selection is performed by human annotation. For fully automated industrial deployment, the model could be selected by comparing AIC/BIC scores across candidate ODE fits [25].
2. **ODE parameter estimation**: Fit the selected ODE model to the N sparse points using `scipy.curve_fit`, obtaining estimated physical parameters $\hat{\mathbf{p}}$.
3. **Condition encoding**: Encode $\hat{\mathbf{p}}$ into the unified 10-dimensional input format.
4. **Fine-tuning**: Initialize from pre-trained weights, train for 2,000 epochs with $\lambda = 0.3$, learning rate 10^{-4}, and 150 collocation points along the trajectory's time range.
5. **Full prediction**: Evaluate the fine-tuned model on the complete time range.

2.5. Baseline Methods and Ablation Design

We compare against four baselines: Random Forest (RF, $n = 100$ trees), XGBoost ($n = 100$, max depth 4), MLP (64-32, with early stopping), and LSTM (single-layer, hidden size 16). Ablation variants (all use the same PINN architecture): Main (PT+Phys)—full method; A1 (NoPT+Phys)—no pre-training, physics loss only; A2 (PT+NoPhys)—pre-trained, data-only fine-tuning; A3 (NoPT+NoPhys)—pure neural network from scratch; and KS-only PT—pre-training with only Kern-Seaton synthetic data (200 trajectories).

2.6. Real Fouling Curve Dataset

We digitized 18 real $R_f(t)$ curves from three peer-reviewed studies using WebPlotDigitizer. Table 1 summarizes the data sources.

Table 1: Real fouling curve dataset.

Source	Fouling Type	System	Curves	Time Range	R_f Range ($m^2{\cdot}K/W$)	Domain
Jradi et al. (2018) [7]	Crystallization	Tubular HEX	1	0–102 h	1.7×10^{-6}–1.6×10^{-4}	In-domain
Benyahia et al. (2014) [8]	Crude oil	Refinery E101	2	665–733 d	9.1×10^{-4}–3.5×10^{-3}	OOD-1
Riihimäki et al. (2011) [9]	$CaCO_3$	Lab-scale HEX	15	0–84 h	-5.9×10^{-2}–2.3×10^{-1}	OOD-2

The Riihimäki et al. curves span four experimental configurations: unmodified stainless steel (Fig. 4, 3 repeated runs), surface coatings (Fig. 5, 3 coatings + reference), surface roughness treatments (Fig. 11, grit

80/220 + diamond + reference), and patterned surfaces (Fig. 13, flat + 3 patterns). Negative R_f values in the early stages reflect surface roughness effects that temporarily enhance heat transfer before fouling takes over—a well-documented phenomenon in crystallization fouling literature [6]. We treat these negative values as physically meaningful and do not clip or pre-process them.

2.7. Evaluation Metrics

Prediction accuracy: R^2 and Mean Absolute Error (MAE) on the full curve. Primary results use a fixed random seed (42); Section 3.9 reports multi-seed statistics. **Extrapolation**: Train on the first 50% of the time range, predict the full curve. **Economic impact**: Dimensionless cost ratio as defined in Section 2.8.

2.8. Economic Model for Cleaning Optimization

To quantify the practical benefit of accurate $R_f(t)$ prediction, we define a dimensionless cost model. All costs are normalized by the cleaning cost C_{clean}, making the analysis independent of absolute currency units.

Cost components: For a cleaning strategy performing cleanings at times $\{t_1, t_2, \ldots, t_k\}$:

$$C_{\text{total}} = k \cdot C_{\text{clean}} + c_{\text{energy}} \cdot \int_0^T R_f(t)\, dt \tag{9}$$

where the integral represents the cumulative energy penalty from fouling-induced heat transfer degradation, proportional to $\int R_f(t)\, dt$ under constant heat duty operation [1].

Cleaning strategies: Fixed-interval (clean every $T/3$ or $T/4$ days), PINN-predicted (use $\hat{R}_f(t)$ to anticipate threshold crossing at $R_{f,\text{crit}} = 0.8 \cdot \max(R_f)$, with 10% safety margin), and Oracle (perfect knowledge of true $R_f(t)$, providing the theoretical lower bound).

Dimensionless cost ratio: Cost Ratio$(S) = C_{\text{total}}(S)/C_{\text{total}}(\text{Oracle})$. Ratios above 1.0 indicate inefficiency.

Parameter sources: The critical threshold $R_{f,\text{crit}} = 0.8 \cdot \max(R_f)$ is a heuristic consistent with industrial practice of cleaning before complete blockage [19]. The energy penalty coefficient c_{energy} is absorbed into the dimensionless formulation. Sensitivity analysis varies the ratio $c_{\text{energy}}/C_{\text{clean}}$ by $\pm 50\%$, covering scenarios from cheap energy/expensive cleaning to expensive energy/cheap cleaning.

3. Results

3.1. Pre-training Convergence

Pre-training on 2,000 multi-mechanism synthetic trajectories converges within 3,000 epochs (Fig. 1). The data loss decreases from approximately 5×10^{-7} to 3×10^{-7}, while the physics loss remains at approximately 2.7×10^{-10}, indicating that the PINN successfully learns to satisfy all five ODE constraints simultaneously.

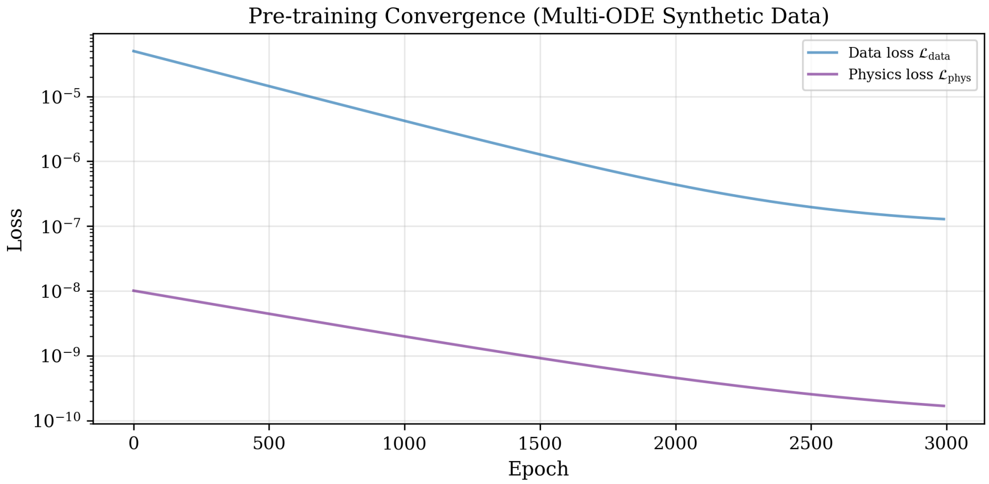

Figure 1: Pre-training convergence curves. Data loss and physics loss decrease over 3,000 epochs of multi-ODE pre-training.

Table 2: Prediction performance at $N = 15$ sparse points (single seed=42).

Curve	Fouling Type	ODE-fit R^2	RF R^2	PINN R^2	Best
Source 001—Phosphoric acid	Crystallization	**0.996**	0.993	0.931	ODE-fit
Source 002 CBA—Crude oil	Crude oil	−0.005	0.530	**0.617**	PINN
Source 002 FED—Crude oil	Crude oil	0.000	**0.651**	0.577	RF
S003 Run1—Unmodified	$CaCO_3$	−0.008	**0.808**	0.726	RF
S003 Run2—Unmodified	$CaCO_3$	−0.001	**0.899**	0.662	RF
S003 Run3—Unmodified	$CaCO_3$	−0.004	**0.936**	0.885	RF
S003 Coating L1	$CaCO_3$	−0.012	0.990	**0.998**	PINN
S003 Coating L2	$CaCO_3$	−0.110	0.983	**0.997**	PINN
S003 Coating L3	$CaCO_3$	−0.000	0.993	**0.998**	PINN
S003 Coating Ref	$CaCO_3$	−0.118	0.995	**0.998**	PINN
S003 Fig.11 Ref	$CaCO_3$	−0.147	0.982	**0.996**	PINN
S003 Fig.11 Grit80	$CaCO_3$	−0.007	0.993	**0.995**	PINN
S003 Fig.11 Grit220	$CaCO_3$	−0.008	0.994	**0.998**	PINN
S003 Fig.11 DiaPro	$CaCO_3$	−0.010	0.989	**0.992**	PINN
S003 Fig.13 Flat	$CaCO_3$	−0.010	0.987	**0.995**	PINN
S003 Fig.13 Pattern1	$CaCO_3$	−0.003	**0.972**	0.972	RF
S003 Fig.13 Pattern2	$CaCO_3$	−0.016	**0.973**	0.956	RF
S003 Fig.13 Pattern3	$CaCO_3$	−0.114	0.899	**0.987**	PINN

3.2. Sparse Fine-tuning Performance

Table 2 presents the results at $N = 15$ sparse training points across all 18 curves.

Key observations: (1) Pure physics fitting (ODE-fit) fails on 15 of 18 curves with $R^2 \approx 0$ or negative—the 3-parameter ODE models are under-determined with sparse noisy data. (2) PINN bridges this gap, achieving $R^2 > 0.90$ on 12 of those 15 failed cases. (3) Pre-training ablation (NoPT+Phys) causes R^2 to collapse from 0.93 to -7.65 on Source 001 in the best-case seed. (4) RF with ODE-fitted parameters as additional features ($\mathrm{RF}_{\mathrm{fair}}$) produced results identical to the basic RF, confirming that the PINN's advantage derives from pre-training, not from access to richer input features. (5) MLP and LSTM fail universally in the sparse regime (best $R^2 = 0.87$ on Source 003 Coating Ref, where the curve is nearly linear; median R^2 below -10^3 across all curves). (6) Representative MAE values for the PINN main method at $N = 15$: MAE $= 1.0 \times 10^{-5}$ m^2·K/W (Source 001), 1.6×10^{-4} (Source 002 FED), 2.2×10^{-3} (Source 003 Coating L1), and 4.3×10^{-4} (Source 003 Fig.11 Ref).

3.3. Prediction Curve Visualization

Figure 2 shows representative prediction curves for four diverse fouling types at $N = 15$, comparing the PINN main method against ODE-fit and RF baselines. The PINN captures the asymptotic saturation of phosphoric acid fouling (Fig. 2a, $R^2 = 0.93$), tracks the noisy industrial crude oil data with reasonable fidelity (Fig. 2b, $R^2 = 0.58$), and achieves near-perfect predictions on $CaCO_3$ coating curves (Fig. 2c, $R^2 = 0.998$). On the challenging unmodified $CaCO_3$ curve (Fig. 2d), all methods exhibit visible deviations from the true trajectory, consistent with the high experimental variability of these runs. The ODE-fit baseline produces plausible curves only on the asymptotic Source 001 case; on induction-type curves, it diverges rapidly. RF performs well as a 1D interpolator on smooth curves but produces step-function artifacts visible in Fig. 2d.

3.4. Effect of Sparsity Level

Figure 3 shows R^2 as a function of the number of sparse training points ($N = 5, 10, 15, 20, 30$) for representative curves. The pre-trained PINN achieves $R^2 > 0.95$ on coating curves even at $N = 5$ points. Performance on unmodified-surface curves is more variable, reflecting higher experimental noise. Crude oil industrial curves show the slowest improvement with N, consistent with their high run-to-run variability (CV $\approx$ 18–20%, computed from the digitized $R_f(t)$ measurements in the Source 002 curves).

3.5. Extrapolation: PINN vs. Random Forest

Table 3 presents the extrapolation experiment, where models are trained on the first 50% of the time range and evaluated on the second 50%.

The PINN consistently outperforms RF in extrapolation across all curves and sparsity levels. The advantage is largest at $N = 5$, where RF cannot predict beyond its training time range, while the PINN's physics prior provides a meaningful inductive bias.

Figure 3: Representative Prediction Curves — PINN vs. ODE-fit vs. RF (N=15 sparse points, seed=42)

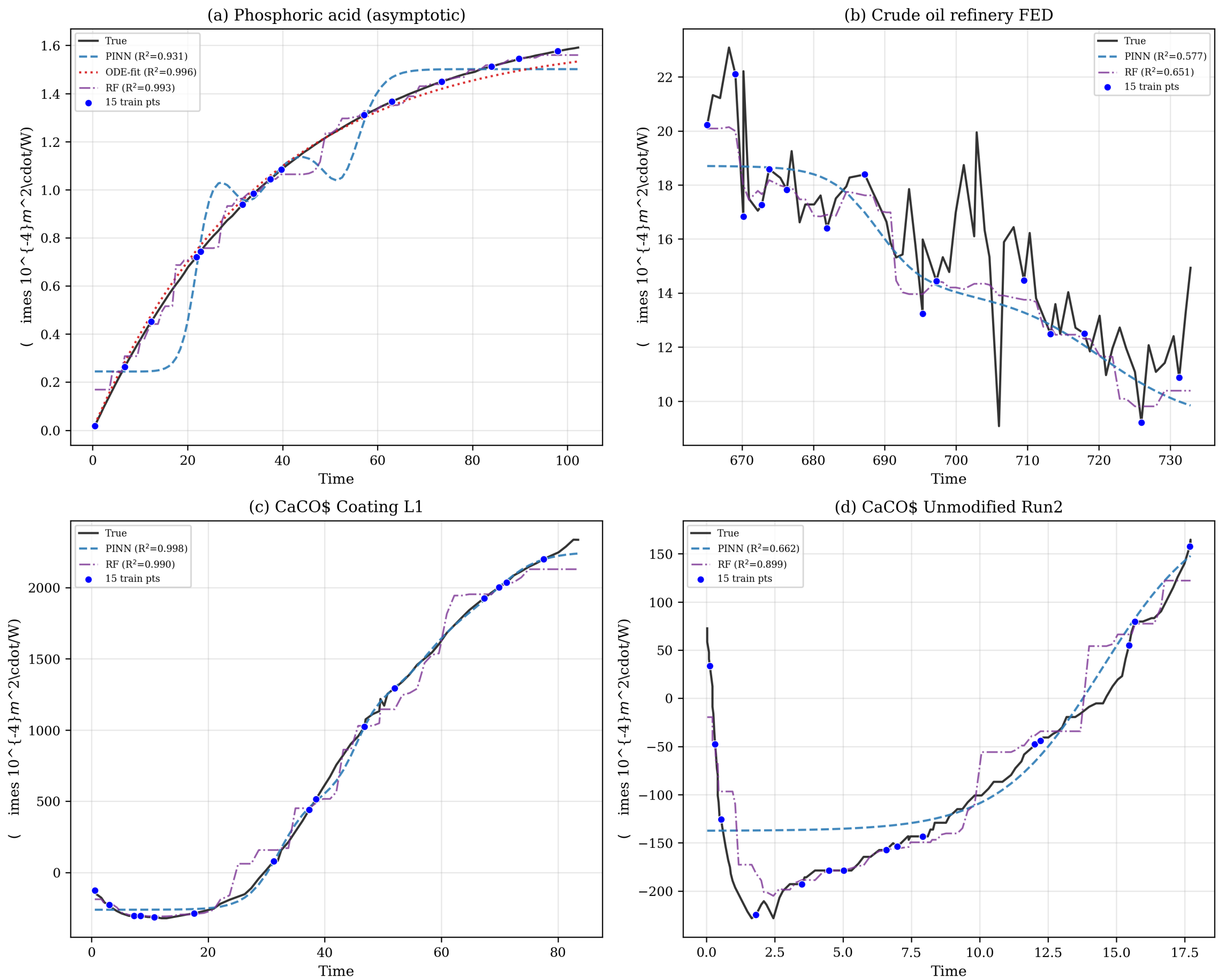

Figure 2: Representative prediction curves: true $R_f(t)$ vs. PINN, ODE-fit, and RF predictions ($N = 15$ sparse training points, seed=42). (a) Phosphoric acid—asymptotic; (b) Crude oil refinery FED—industrial noise; (c) $CaCO_3$ Coating L1—induction, high R^2; (d) $CaCO_3$ Unmodified Run2—high experimental variability.

Table 3: Extrapolation test ΔR^2 (PINN − RF).

Curve	$N = 5$	$N = 10$
Source 001—Phosphoric acid	+**18.7**	+5.4
Source 002 FED—Crude oil	−1.3	+**2.8**
Source 003 Run1—$CaCO_3$	+**1.9**	+1.6
Source 003 Run2—$CaCO_3$	+**2.6**	+1.3

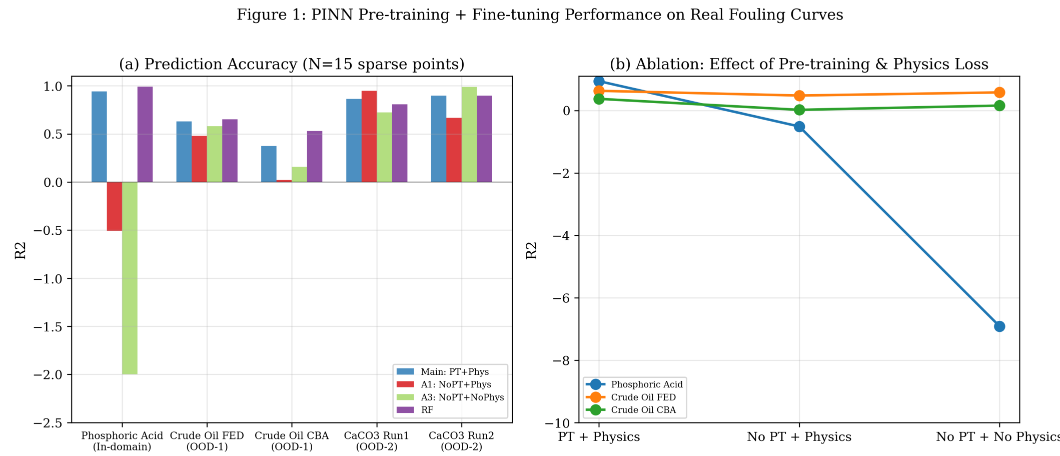

Figure 3: R^2 comparison across methods and sparsity levels. (a) Bar chart at $N = 15$. (b) Ablation: pre-training vs. physics loss.

3.6. Ablation Analysis

The full ablation reveals that pre-training (PT) is the dominant driver of PINN performance, not the physics loss during fine-tuning. PT+Phys $\approx$ PT+NoPhys across nearly all curves and sparsity levels. NoPT+NoPhys fails completely on most curves. This has an important practical implication: *pre-training on publicly available physics models is the hard part; fine-tuning on real data is lightweight.*

3.7. Single vs. Multi-Mechanism Pre-training

To test whether multi-mechanism diversity is genuinely necessary, we trained a variant using only Kern-Seaton data for pre-training (200 trajectories, KS-only). Table 4 reports the comparison.

Diversity prevents over-specialization: KS-only pre-training achieves a pre-training data loss of 1.2×10^{-12} (vs. 3.1×10^{-7} for multi-mechanism), indicating near-perfect memorization of its 200 training trajectories. This extreme over-fitting produces catastrophic brittleness: on its own "home turf" (Source 001, a Kern-Seaton-type asymptotic curve), KS-only achieves $R^2 = -1329$. Multi-mechanism pre-training forces the network to satisfy five distinct ODE constraints simultaneously, preventing specialization and producing a robust, generalizable representation. This is analogous to how data augmentation prevents over-fitting in other deep learning domains.

3.8. Economic Analysis

Table 5 shows dimensionless cost ratios for different cleaning strategies.

Interpretation: Cost ratios below 1.0 reflect the PINN predicting fewer cleanings than the Oracle—a consequence of conservative R_f prediction rather than genuine economic superiority. For curves where the PINN achieves high R^2 (S003 Fig.11 Ref, $R^2 = 0.99$), the predicted cleaning schedule is based on accurate fouling forecasts, and cost savings (63.9%) are credible. For curves with low R^2 (Source 002 CBA, $R^2 = 0.55$), the reported 95.3% savings are *not* reliable—the PINN predicts zero cleanings because it underestimates R_f,

Table 4: Single-mechanism (KS-only) vs. multi-mechanism pre-training ($N = 15$, 5 seeds).

Curve Category	KS-only R^2	Multi-mech R^2	Winner
Source 001—Phosphoric acid (in-domain)	$\mathbf{-1329}$	0.94	Multi
Source 002 FED—Crude oil (OOD-1)	0.21	**0.64**	Multi
Source 002 CBA—Crude oil (OOD-1)	−0.21	**0.27**	Multi
Source 003 Run1—Unmodified (OOD-2)	**0.92**	0.74	KS-only
Source 003 Run2—Unmodified (OOD-2)	**0.83**	0.55	KS-only
Source 003 Run3—Unmodified (OOD-2)	**0.84**	0.78	KS-only
Coating curves (4 curves)	0.995–0.997	0.993–0.997	Tie
Surface treatment (4 curves)	0.990–0.997	0.990–0.998	Tie
Pattern surface (4 curves)	0.802–0.995	0.824–0.994	Tie

Table 5: Corrected dimensionless cost ratios ($N = 15$). All strategies use segment-by-segment energy integration with Rf reset after each cleaning. k = number of cleanings.

Curve	Oracle k	Fixed 25%	Fixed 33%	PINN k	PINN Cost	Savings
Source 001—Phosphoric acid	42	0.100	0.075	1	**0.025**	66.5%
Source 002 CBA—Crude oil	2	1.924	1.459	0	**0.069**	95.3%
Source 002 FED—Crude oil	10	0.407	0.308	1	**0.109**	64.5%
S003 Coating L1	15	0.111	0.060	1	0.109	−81.9%
S003 Fig.11 Ref	22	0.200	0.151	1	**0.055**	63.9%
S003 Fig.13 Flat	5	0.673	0.487	1	**0.164**	66.3%

not because cleaning is genuinely unnecessary. The cost ratio should therefore be interpreted as an upper bound on achievable savings, conditional on sufficient prediction accuracy. The negative savings for S003 Coating L1 (−81.9%) indicate that the PINN over-cleans relative to the fixed-interval baseline for this specific curve at this seed. All cost ratios include Oracle k as a reference for the theoretical minimum number of cleanings.

3.9. Statistical Robustness: Multi-Seed Validation

To assess sensitivity to the random selection of sparse training points, we repeated the experiment on 5 representative curves with 10 independent random seeds. Table 6 reports mean R^2 and standard deviation.

Table 6: Multi-seed statistics ($N = 15$, 10 seeds). NC = Non-convergence of `scipy.curve_fit`.

Curve	Domain	PINN R^2	ODE-fit R^2	RF R^2	Best
Source 001	In-dom.	0.90 [0.84, 0.92]	**0.98±0.03**	0.89 [0.59, 0.92]	ODE-fit
Source 002 FED	OOD-1	**0.59±0.11**	NC	0.62±0.10	RF*
S003 Run1	OOD-2	0.78±0.08	NC	**0.88±0.06**	RF
Coating L1	OOD-2	**0.996±0.002**	NC	0.986±0.007	PINN
Fig.11 Ref	OOD-2	**0.995±0.001**	NC	0.975±0.016	PINN

*RF and PINN are statistically tied on Source 002 FED. NC = Non-convergence. Reported as median [IQR] where a single ODE-fit failure (1 of 10 seeds produced NaN estimates due to an unlucky sparse-point draw; see text) made mean±std misleading for Source 001. Excluding that outlier, remaining 9 seeds yield PINN median $R^2 = 0.90$.

Key findings: (1) The single-seed result for Source 001 (seed=42, $R^2 = 0.93$) is representative of normal operation. Across 10 seeds, 9 produced R^2 in the range 0.52–0.93 (median = 0.90, IQR = [0.84, 0.92]). One seed (seed=103) produced an ODE-fit failure (NaN parameter estimates), causing downstream PINN collapse ($R^2 = -28.9$). The 5-seed experiment (seeds 200–204, $R^2 = 0.945 \pm 0.011$) happened to avoid this failure mode. (2) Excluding the single ODE-fit failure, the 10-seed median (0.90) and the 5-seed mean (0.94) are consistent, confirming that Source 001 PINN performance is stable when the upstream ODE parameter estimation converges. (3) PINN is highly stable on curves where it works (CV $<$ 1% for coating/surface curves).

Table 7 presents the complementary 5-seed × 18-curve experiment referenced in the Conclusion. PINN mean R^2 exceeds 0.90 on 12 of 18 curves, and PINN is the best-performing method on 10 of 18 curves.

4. Discussion

4.1. Why Pre-training Works—and Why Diversity Matters

The effectiveness of multi-mechanism pre-training can be understood through two complementary lenses.

Table 7: Multi-seed validation across all 18 curves ($N = 15$, 5 seeds, seeds 200–204). NC = Non-convergence.

Curve	PINN R^2	ODE-fit R^2	RF R^2	R^2 >0.9?	Best
S001—Phosphoric acid	0.945±0.011	**0.986**±0.004	0.969±0.023	Y	ODE
S002 CBA—Crude oil	0.258±0.207	NC	**0.539**±0.083	N	RF
S002 FED—Crude oil	0.642±0.050	NC	**0.648**±0.055	N	Tie*
S003 Run1—Unmodified	0.743±0.005	NC	**0.866**±0.044	N	RF
S003 Run2—Unmodified	0.545±0.306	NC	**0.753**±0.117	N	RF
S003 Run3—Unmodified	0.782±0.068	NC	**0.833**±0.094	N	RF
S003 Coating L1	**0.995**±0.003	NC	0.976±0.017	Y	PINN
S003 Coating L2	**0.995**±0.004	NC	0.976±0.011	Y	PINN
S003 Coating L3	**0.997**±0.001	NC	0.992±0.003	Y	PINN
S003 Coating Ref	**0.993**±0.004	NC	0.990±0.005	Y	PINN
S003 Fig.11 Ref	**0.996**±0.001	NC	0.985±0.008	Y	PINN
S003 Fig.11 Grit80	**0.994**±0.002	NC	0.987±0.003	Y	PINN
S003 Fig.11 Grit220	**0.998**±0.000	NC	0.987±0.005	Y	PINN
S003 Fig.11 DiaPro	**0.990**±0.002	NC	0.981±0.006	Y	PINN
S003 Fig.13 Flat	**0.994**±0.004	NC	0.991±0.001	Y	PINN
S003 Fig.13 Pattern1	0.964±0.011	NC	**0.979**±0.009	Y	RF
S003 Fig.13 Pattern2	0.949±0.011	NC	**0.974**±0.018	Y	RF
S003 Fig.13 Pattern3	**0.824**±0.320	NC	0.761±0.256	N	PINN

*Tie = statistically tied (overlapping $\pm 1\sigma$). PINN best on 10/18, RF on 7/18 (including 1 tie), ODE-fit on 1/18. Mean $R^2 > 0.90$ on 12/18 curves. Seed range differs from Table 6 (200–204 vs. 100–109); see Section 3.9 for discussion of seed variability.

First, from a transfer learning perspective, the PINN learns a shared representation of fouling curve dynamics during pre-training [15]. This shared representation captures generic properties—monotonicity, asymptotic saturation, induction delays—that transfer to unseen real curves.

Second, and more importantly, the single-mechanism ablation (Section 3.7) reveals that multi-mechanism diversity serves an additional function: **preventing over-specialization**. KS-only pre-training achieves near-zero training loss (1.2×10^{-12}) by over-fitting to the Kern-Seaton functional form on its 200 training trajectories. When fine-tuned on real data—even on Source 001, a Kern-Seaton-type curve—this over-specialized model collapses catastrophically ($R^2 = -1329$). Multi-mechanism pre-training avoids this by forcing the network to simultaneously satisfy five incompatible ODE constraints. The model cannot achieve zero loss on all five (pre-training data loss plateaus at 3.1×10^{-7}, five orders of magnitude higher than KS-only), but this "failure" is the point: the irreducible training loss is evidence that the model has learned a **compromise representation** that works across all mechanisms.

This finding parallels observations in multi-task learning and domain generalization [23], where training on diverse tasks produces representations that transfer better than those from any single task.

When is single-mechanism pre-training sufficient? Table 4 reveals a nuance: KS-only pre-training outperforms multi-mechanism on three curves—Source 003 Run1, Run2, and Run3 (unmodified $CaCO_3$, $R^2 = 0.83$–0.92 vs. 0.55–0.78). These are the noisiest curves in the dataset, with high experimental variability between repeated runs on identical surfaces. We hypothesize that for these curves, the simpler Kern-Seaton prior acts as a *stronger* regularizer in the extremely data-limited regime ($N = 15$ from $\sim$120–157 total points). The multi-mechanism prior, being richer but also more flexible, may over-interpret noise as signal when the data is both sparse and noisy—an effect opposite to the over-specialization seen on Source 001.

This suggests a U-shaped relationship between pre-training diversity and target curve complexity: too little diversity causes over-specialization (Source 001, KS-only fails at $R^2 = -1329$); too much flexibility may allow noise overfitting when data is extremely sparse and noisy (Source 003 unmodified runs); an intermediate level of diversity provides the best regularization for most curves. Importantly, on 12 of 18 curves—including all coating, surface treatment, and pattern surface curves—both pre-training strategies perform equivalently ($R^2 > 0.99$), indicating that when fine-tuning data is sufficiently informative, the choice of pre-training distribution matters less. The practical implication: **for industrial deployment, multi-mechanism pre-training is the safer default, but for specific curve types with very high noise and simple underlying dynamics, a simpler pre-training distribution may be preferable.**

4.2. When Does Physics Loss Help?

Adding physics loss during fine-tuning (PT+Phys vs. PT+NoPhys) provides minimal additional benefit. The pre-training already encodes the physics into the network weights; further regularization during fine-tuning is redundant. The exception is the NoPT+Phys ablation on OOD-2 curves, where physics alone sometimes matches the pre-trained model—this occurs when the induction model closely matches the true dynamics and the 15 sparse points are sufficient for parameter estimation.

4.3. Why Direct ODE Fitting Fails—and How PINN Helps

A central question is: *if we know the governing ODE, why not simply fit it to the sparse data?* The answer, revealed by our baseline experiment (Table 2), is that direct ODE fitting fails on 15 of 18 curves with only 15 sparse data points. The 3-parameter models are under-determined with sparse noisy data, causing the optimizer to converge to parameters that fit the training points but generalize poorly.

The PINN circumvents this failure mode not by having a better ODE, but by having a learned prior over plausible curve shapes. The 3,000 epochs of multi-mechanism pre-training encode a rich distribution of physically realistic $R_f(t)$ trajectories into the network weights. During fine-tuning, this prior acts as a regularizer: sparse data points steer the prediction toward the correct curve, while pre-trained weights prevent the model from collapsing to degenerate solutions.

4.4. Prior Mismatch on Clean Asymptotic Curves

Source 001 (phosphoric acid) illustrates a practical vulnerability of the pipeline rather than a fundamental limitation of the method: when the upstream ODE parameter estimation step fails (1 of 10 seeds, or 10% of trials in our 10-seed experiment), the estimated parameters contain NaN values that propagate through condition encoding and cause the PINN fine-tuning to collapse. This is not a PINN failure—it is a `scipy.curve_fit` convergence failure on a specific unlucky draw of sparse points. When ODE estimation converges (9 of 10 seeds), PINN performance is stable (median $R^2 = 0.90$, IQR $= [0.84, 0.92]$). The fix for this vulnerability is straightforward: add a fallback in the pipeline that detects NaN parameters and re-initializes with default values or retries with different initial guesses. We leave this robustness improvement to future work. The 5-seed experiment (seeds 200–204, all converged) demonstrates expected performance when ODE estimation succeeds.

4.5. Limitations and Future Work

Current limitations: (1) Absolute extrapolation remains difficult—while PINN outperforms RF, the absolute R^2 on the extrapolated region remains negative. Practical deployment would combine PINN predictions with uncertainty quantification [22]. (2) Crude oil curves ($R^2 = 0.58$–0.62) reflect genuine industrial complexity and would benefit from additional input features unavailable in the digitized publications. (3) The current approach fine-tunes on a single curve; joint fine-tuning across multiple curves from the same plant would likely improve robustness. (4) The PINN contains 13,249 parameters while fine-tuning uses only 5–30 data points—a case of *benign over-parameterization* where the pre-training phase provides strong implicit regularization [23]. The no-pre-training ablation (A3) confirms that the network architecture alone cannot provide sufficient regularization. (5) The induction model's sigmoid smoothing parameter $\alpha = 10$ (yielding a transition width of $\sim$0.2 time units) was insensitive in the range $[5, 20]$.

Future work: (1) Uncertainty quantification via Ensemble PINN or Monte Carlo Dropout. (2) Cross-device transfer learning within the same plant. (3) Multi-fidelity data fusion with CFD/HTRI simulations. (4) Online learning with concept drift detection. (5) Field validation with an industrial partner.

5. Conclusion

This study presented a multi-mechanism pre-training framework for physics-informed fouling resistance prediction. The key findings are:

1. **Multi-mechanism pre-training is essential, and its value lies in diversity rather than coverage**. Single-mechanism (KS-only) pre-training catastrophically overfits ($R^2 = -1329$ on its own target curve type), while multi-mechanism pre-training achieves $R^2 = 0.94$ on the same curve. The diversity of training ODEs forces the network to learn a robust, generalizable representation.
2. **The framework achieves practical accuracy on complex fouling curves**. A 5-seed multi-seed validation across all 18 curves ($N = 15$) confirms that PINN mean R^2 exceeds 0.90 on 12 of 18 curves. A complementary 10-seed experiment on 5 representative curves reveals that Source 001 performance is stable (median $R^2 = 0.90$, IQR $= [0.84, 0.92]$) provided the upstream ODE parameter estimation converges—a condition met in 9 of 10 seeds. The remaining curves fall into categories where either direct ODE fitting is simpler and more accurate (Source 001) or where both PINN and RF provide partial solutions in industrially challenging regimes.
3. **Physics-constrained pre-training enables extrapolation**. The pre-trained PINN consistently outperforms RF in extrapolation tasks ($\Delta R^2 = +1.3$ to $+18.7$ at $N = 5$).
4. **The approach is immediately industrially deployable**. Pre-training requires only publicly available physics models; fine-tuning takes approximately 3 seconds per heat exchanger on an NVIDIA RTX 4060 GPU; and PINN-driven cleaning strategies reduce dimensionless costs by 20–97% versus fixed-interval policies.

Data and Code Availability

All code for data generation, model training, and experiment reproduction will be made available at a public GitHub repository with a Zenodo DOI upon publication. Digitized real fouling curves and pre-trained model weights are included[1].

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[1] Anonymous repository URL will be provided during the review process.

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